EPPS 6302 Methods of Data Collection and Production: Assignment 7

Social media data from the YouTube API

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> **What you will produce**
>
> A small, documented collection of YouTube videos and comments on the 2026 US midterm elections — made with an API key and a quota plan, before and after Election Day (3 November); a quanteda analysis of the comments by channel; and a short report on what a comment section is a sample of, compared with the tweets and presidential documents of Assignment 6.

0. Set up

a. A YouTube Data API key in your user `.Renviron` as `YOUTUBE_KEY=...` (Prepare for Class 9). Check with `nzchar(Sys.getenv("YOUTUBE_KEY"))`. If it is `FALSE`, run the checks in Part 1b of the script — a `.Renviron` in your project folder replaces `~/.Renviron`.

b. The development version of *tuber* (2.0.0.9000), with a current *httr2*: `pak::pak(c("soodoku/tuber", "httr2"))`. Check `packageVersion("tuber")` and `packageVersion("httr2")` (1.1.0 or later). Old tutorials use `get_comment_threads()`, `get_channel_stats()` and `get_stats()`; all three were renamed (Part 0).

c. Download `YouTubeAPI02_usmidtermelections.R` from Teams. The parts below follow it. An API key is all this assignment needs: **skip Part 7** (OAuth).

d. `data/` and `.Renviron` in `.gitignore` — check before the first call.

1. Read

a. *tuber* — the vignettes *API conventions* and *Batch processing & quota management*, https://gojiplus.github.io/tuber/.

b. YouTube Data API **quota costs**, https://developers.google.com/youtube/v3/determine_quota_cost, and the **Developer Policies**, sections III.E.4 and III.E.6, https://developers.google.com/youtube/terms/developer-policies.

c. Freelon, Deen. 2018. "Computational Research in the Post-API Age." *Political Communication* 35 (4): 665–668.

2. Collect

a. Plan the quota first (Part 2). Before any call, write a table: each call you intend to make, how many pages, and its cost. Search calls (100 a day) and units (10,000 a day) are separate budgets. After collecting, compare with `yt_get_quota_usage()`.

b. Find the videos — twice (Part 3a). Search `"US midterm election"` for videos published 1 September – 27 October 2026, once with `order = "relevance"` and once with `order = "viewCount"` (100 results each).

i. How many videos appear in both lists? Which channels dominate each?
ii. `attr(x, "total_results")` against `nrow(x)`: what does YouTube say exists, and what did it give you?
iii. Search can return the same video more than once. Report `sum(duplicated(x$video_id))` for each list, and keep **one row per video** (`distinct(video_id, .keep_all = TRUE)`) before joining the video statistics of Part 4.

c. Choose three channels that covered the campaign and differ in kind — for example a national network (NBC News, as in Part 3b), a cable news channel, and a local station or an international broadcaster. For each: `get_channel_details()`, then find its election videos with `list_channel_videos()` (the cheap route, Part 3b) rather than search, and filter titles with a pattern you write down (e.g. `"midterm|election|senate|house race|ballot"`).

i. Did the channel route find videos the search missed?
ii. The uploads list runs newest first, 50 videos per unit. How many units did it take to reach 1 September for each channel?
iii. Is your title filter too narrow or too broad? Read 20 titles it kept and 20 it dropped.

d. Collect comments with `collect_comments()` (Part 5c) from **two to four videos per channel**, at most 300 comments per video. Write the collection log and the manifest (Part 5d). Record for each video the comment count it reports (`get_video_stats()`) and the number you received.

i. Before you join comments to video titles, make sure each video appears once and each comment once: `distinct(comment_id, .keep_all = TRUE)` and `left_join(..., relationship = "many-to-one")`. A repeated video otherwise copies its comments, and `corpus()` stops with *docnames must be unique*.
ii. Check the manifest records the search term **exactly** as you sent it — and that the key is nowhere in it.

e. After Election Day. Repeat 2b for videos published from 4 November 2026 to the day you collect, with the same term and both orderings, and collect comments from the same three channels' post-election videos. Use a new manifest with its own `delete_by` date.

> **The data expire.** Comments collected with an API key are *non-authorized data*: the YouTube API Developer Policies (III.E.4.d) allow you to keep them for no more than 30 days. Write a `delete_by` date into each manifest, delete `data/youtube/` by then, and make your report reproducible from the code and the manifest — not from a saved copy. Scraping YouTube pages or transcripts outside the API is not permitted (III.E.6); do not use `yt-dlp` or transcript scrapers for this assignment.

3. Analyze

a. Describe the collection before the text.

i. Comments received vs. comments reported, per video. Where did the difference go (held for review, deleted, replies, your cap)?
ii. `count(status)` on your log: how many videos were `ok`, `disabled`, `none`, `error`? Which channels disable comments?
iii. Share of replies; distribution of `like_count` — what share of all likes goes to the top 1% of comments?

iv. How concentrated is authorship: what share of comments comes from the 10% most active authors? Exact duplicates? (Report counts, not names.)

b. Read, then count (quanteda; Part 6 of the script).

i. A corpus with docvars `channel_title`, `video_id`, `is_reply`, `like_count`, `date`, and a `period` (before / after Election Day). Say which unit you chose — comment or thread — and what you did with emoji, URLs and mentions.
ii. `kwic()` for *democrat**, *republican**, *trump*, *vote**, *ballot**. Read 20 lines for each before any model, and tally what the words are doing: naming a party, a candidate, the process, a grievance?
iii. Phrases as features: `tokens_compound()` for *house majority*, *senate race*, *early voting*, *mail in*, *election fraud* — and add two phrases of your own from your `kwic()` reading.
iv. Keyness: one channel's comments against the other two. Then swap the target. Then compare **before** with **after** Election Day. What is distinctive, and is it about the election or about the audience?
v. Sentiment by channel and period with `data_dictionary_LSD2015` and `dfm_lookup()`. **Validate it**: hand-code 40 randomly sampled comments as positive, negative or neither, and report agreement with the dictionary. Where does a dictionary misread election talk (sarcasm, "landslide", "rigged")?

c. Compare with Assignment 6. Biden–Xi 2021 tweets; presidential documents mentioning Xi; midterm 2026 comments. For each: who produced the text, for whom, what selected it, whether it can be collected again, and what kind of claim it can support — about the public, an audience, or a government.

i. *Optional.* Reply network with *vosonSML* (Part 9): who replies to whom, and is the conversation one crowd or several?
ii. *Optional.* Rerun one comment request with *httr2* (Part 8) and confirm you get the same first page. What does *tuber* add?

> **Every comment section has a data-generating process.** Who chose to comment, under which video, on which channel; what the channel moderated and YouTube held back; how `order` ranked what you saw first; when you ran the query — before or after the result was known. In an election season, add who comments *many* times and what repeated text may be. Say which in your report — it decides whether your keyness and sentiment results describe *voters*, *an audience*, or *the comments YouTube showed you*.

4. Submit

a. Post `assign07.qmd` under **Assignments** on your website, with the plots, the quota table, a short report (about 700 words), and the **data note** for the YouTube collection: source; permission (API terms, key, quota, 30-day retention); method (search term, windows, `order`, channels, title filter, caps); selection; known gaps.
b. Post the **collection logs**, the **manifests** and aggregate tables only — **not** the comments, **not** author names or channel IDs of commenters, and **not** your key. Quote at most a few short comments, without their authors, and do not quote claims about voting or fraud without saying what they are.